QP CODE: 25901338

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

Discipline Specific Core Course (DSC) –MG4DSCHPY200

Applied physiology

(2024 ADMISSION ONWARDS)

Duration: 1Hour 30 minutes Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple ChoiceQuestions

Answer all questions

Each question carries 1 mark

1. b) Hypervolemia
2. c) Removal of dust particles
3. c) Proteins
4. b) Hepatic stage
5. d) Increases
6. c) Liver

Part B

Short Answer Questions

Answer any **2** questions.

Each question carries **2** marks.

7. **Explain the Role of chemoreceptors in exercise-induced hyperventilation.**

- Chemoreceptors detect changes in CO_2, O_2, and pH during exercise. (1 Mark)
- Increased CO_2 and decreased pH stimulate chemoreceptors, which signal the respiratory center to increase the rate and depth of breathing (hyperventilation). (1 mark)

8. **Define alkalosis:**

- Alkalosis is a condition in which the blood pH rises above normal (>7.45) due to

decreased hydrogen ion concentration.(2 marks)

9. **Recall complete heart block**

- Complete heart block is the condition in which the impulses produced by SA node cannot reach the ventricles.
- It is also called complete atrioventricular block or third degree heart block.
- The ventricles beat in their own rhythm, independent of atrial beat. It is called idioventricular rhythm.
- Complete heart block occurs due to AV nodal block and infranodal block.

(**1/2 mark for each point)**

10. **Define hemolysis**
    - **Destruction of RBC's (1 mark)**
    - **Leading to the release of haemoglobin into the plasma (1 mark)**

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks.

11. **Discuss about the valves and blood vessels present in human heart.**

Valves of heart

- The auricles of each side communicate with the corresponding ventricle by means of the auriculo-ventricular opening/aperture guarded by AV valves.
- The right AV valve has three cusps and is called tricuspid valve.
- The left one has only two cusps and is called bicuspid or mitral valve.
- These valves open towards the ventricle permitting the flow of blood only from the auricles to ventricles. (**2 marks – ½ mark for each point)**

Blood vessels in heart

- The right auricle receives three important veins namely: superior venacava (precaval vein) from anterior part of the body, inferior venacava (postcaval vein) from the posterior part of body and coronary vein from the heart wall. (**1 mark)**
- The left auricle receives four pulmonary veins, two from right lung and two from left lung. (**1/2 mark)**
- Leading from the right ventricle is the pulmonary artery that carries blood to the lungs. (**1/2 mark)**
- Arising from the left ventricle is the aorta whose branches supply blood to the different parts of the body. (**1/2 mark)**
- At the opening of each of these is a set of three semilunar valves. (**1/2 mark)**

12. **Describe Effects of exposure to severe cold:**

- Death Risk – Severe cold can be fatal; survival time decreases with lower temperatures.(1 mark)

- Hypothalamus Failure – Below 34.4°C regulation is affected; below 25°C it stops completely.(1 mark)
- Body Suppression – Metabolism slows, shivering stops, leading to coma.(1 mark)
- Frostbite – Freezing of tissues, especially ears, fingers, and toes.(1 mark)
- Tissue Damage – Prolonged cold causes cell death and gangrene. (1 mark)

13 **Explain renal failure and its complications.**

- Renal failure refers to failure of excretory functions of kidney. It is usually, characterized by decrease in glomerular filtration rate (GFR). **(1 mark)**
- If 50% of the nephrons are affected, GFR decreases only by 20% to 30%. It is because of the compensatory mechanism by the unaffected nephrons. **(1 mark)**

Complications (3 marks)

1. Deficiency of calcitriol resulting in reduction of calcium absorption from intestine and hypocalcemia. Deficiency of calcitriol and hypocalcemia may cause secondary hyperparathyroidism in some patients
2. Deficiency of erythropoietin resulting in anemia
3. Disturbances in acidbase balance.

14. **Effects of exercise on blood and blood volume**

- Mild hypoxia during exercise → ↑ erythropoietin secretion
- Stimulates bone marrow →↑ RBC production from bone marrow. (2 marks)
- ↑ CO_2 levels → ↓ blood pH. (1 mark)
- Sweating causes fluid loss → ↓ blood volume. (1 mark)
- Severe exercise → hemoconcentration and dehydration. (1 mark)

15. Pneumonia and its effects on respiration.

- Pneumonia is an infection of the lungs in which alveoli fill with fluid and pus.
- This leads to reduced gas exchange between air and blood.
- Oxygen supply decreases (hypoxia) and carbon dioxide removal is impaired.
- Causes difficulty in breathing (dyspnea) and rapid respiration.
- In severe cases, it may lead to respiratory failure.(5 marks)

16. Explain heart failure on the basis of side of heart affected.

- **Systolic Heart Failure -** the heart failure due to the decreased ability of heart to contract. It is caused either by muscular weakness or valvular defect. Ventricles may be filled with blood but cannot pump it out with sufficient force. As a result, more amount of blood remains in ventricle, accumulating in lungs or systemic veins or both. **(2 marks)**

- **Diastolic Heart Failure** - the heart failure that occurs when the ventricles cannot relax properly due to the stiffening of cardiac muscle. So, there is reduction in ventricular filling and cardiac output. (**1 mark)**
- **Right sided heart failure** - occurs due to loss of pumping action of the right side of the heart. So, blood accumulates in right atrium and blood vessels. It causes edema in the feet, ankles, legs and abdomen. (**1 mark)**
- **Left Sided Heart Failure** - due to the loss of pumping action of the left side of the heart. It causes congestion of lungs. (**1 mark)**

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

17. Types of apnea & sleep apnea effects

- Types of Apnea (5 marks)
  - ➢ Central apnea – Failure of respiratory center; no breathing effort
  - ➢ Obstructive apnea – Airway blockage despite breathing effort
  - ➢ Mixed apnea – Combination of central and obstructive apnea
- Effects of Sleep Apnea (5 marks)
  - ➢ Repeated stoppage of breathing during sleep
  - ➢ Hypoxia and hypercapnia
  - ➢ Disturbed sleep and daytime sleepiness
  - ➢ Fatigue and morning headache
  - ➢ Long-term: hypertension, cardiac arrhythmias, risk of sudden death

18. **Explain the structure of nephron with the help of a neatly labelled diagram.**

- **Diagram – (**1 mark)

- Each nephron is formed by two parts
  a) A blind end called renal corpuscle or Malpighian corpuscle
  b) A tubular portion called renal tubule. (**1/2 mark)**

  a) RENAL CORPUSCLE **(1/4 mark each – 1 marks)**
- Renal corpuscle or Malpighian corpuscle is a spheroidal and slightly flattened structure with a diameter of about 200 µ.
- It is situated in the cortex of the kidney either near the periphery or near the medulla.
- Function of the renal corpuscle is the filtration of blood which forms the first phase of urine formation.
- It has two parts: glomerulus and Bowman's capsule
- i) **Glomerulus (1/4 mark each – 1 1/2 marks)**

- Glomerulus is a tuft of capillaries enclosed by Bowman capsule. It consists of glomerular capillaries interposed between afferent arteriole on one end and efferent arteriole on the other end.
- Glomerular capillaries arise from the afferent arteriole. After entering the Bowman capsule, the afferent arteriole divides into 4 or 5 large capillaries.
- Each large capillary subdivides into many small capillaries. These small capillaries are arranged in irregular loops. All the smaller capillaries finally reunite to form the efferent arteriole, which leaves the Bowman capsule.
- The afferent vessel is short and wide whereas the efferent arteriole is narrow and long. These arrangements make the glomerular B.P much higher and facilitate filtration.
- Glomerular capillaries are made up of single layer of endothelial cells, which are attached to a basement membrane.
- Endothelium has many pores called fenestrae or filtration pores. These pores helps in the filtration process.

ii) **Bowman's Capsule** (**2 marks – ¼ mark for each point**)

- Bowman's capsule consists of inner visceral layer surrounding the glomerulus and outer parietal layer bordering the renal corpuscle.
- Between these two layers is urinary space where the primary urine is produced.
- Epithelial cells of the visceral layer fuse with the basement membrane but the fusion is not complete.
- Each cell is connected with basement membrane by cytoplasmic extensions of epithelial cells called pedicles or feet.
- Epithelial cells with pedicles are called podocytes.
- The pedicles are arranged in an interdigitating manner leaving small cleft-like spaces in between.
- The cleft like space is called slit pore.
- These filtration slits are components of filtration barrier for for ultrafiltration of blood plasma and thus for primary urine production.

b) RENAL TUBULE (**4 marks – ¼ mark for each point**)

1. Proximal Convoluted Tubule or PCT or Pars Convoluta

- It is about 14 mm long. It forms about 1/3rd of renal tubules. It is lined by large columnar (cuboidal) cells having brush borders or microvilli which are prolongations of cytoplasm.
- It increases the surface area for absorption. Tubular invaginations to cytoplasm between microvilli are called apical canaliculi which help in protein absorption.
- 60-80% of sodium, chloride and water entering the PCT are reabsorbed here.
- The rate of reabsorption is directly proportional to GFR.

2. Henle's Loop (Pars Recta)

- It is a 'U' shaped loop having descending and ascending limbs.
- It is lined with squamous epithelial cells.
- The cells contain many mitochondria, but the numbers of microvilli are less.

- Pars recta contain thick descending segment, thin descending segment, Hairpin bend, thin ascending segment and thick aescending segment.

3. Distal Convoluted Tubule

- These are lined by cubical epithelial cells and has very few or no microvillus.
- There is extensive infoldings of the cell membrane. Basal invaginations also present.
- The absorptive activity of the DCT is under the influence of ADH which determines the volume of urine.
- Distal tubule cells are able to reabsorb small quantities of water and electrolytes due to a smaller number of micrivilli. They also maintain acid- base balance.

4. Collecting Tubule or Straight Tubule (Collecting duct)

- It receives urine from distal tubule. It is lined with high columnar cells.
- It contains two types of cells. Principal (P) cells or light cells are more numerous and have light cytoplasm with few granules, mitochondria and microvilli.
- Intercalcated cells (I cells) or dark cells have dark cytoplasm, many granules and microvilli.
- Several collecting tubules from different nephrons, joins and form the duct of Bellini which opens at the apex of the renal pyramid.

19. **Hemostasis & stages**

Hemostasis is the process of stopping bleeding by forming a clot at the site of blood vessel injury(.1 mark) **Stages of Hemostasis**

i. Vasoconstriction (2marks)

Immediate constriction of injured blood vessels to reduce blood loss; mediated by substances like serotonin released from platelets.

ii. Platelet Plug Formation(2marks)

Platelets adhere to exposed collagen, get activated, and aggregate to form a temporary plug at the injury site.

iii. Coagulation (Blood Clotting) – Steps(5 marks)

- Formation of Prothrombin Activator- Formed by intrinsic (within blood) and extrinsic (tissue injury) pathways.
- Prothrombin → Thrombin- Prothrombin is converted into thrombin in the presence of calcium.
- Fibrinogen → Fibrin- Thrombin converts fibrinogen into fibrin threads.
- Clot Formation- Fibrin threads form a mesh over platelet plug to create a stable clot.
- Clot Stabilization- Factor XIII strengthens fibrin to make the clot firm and permanent.

20. **Explain the distribution, absorption, transport and storage of iron in our body.**

Distribution of Iron in the Body

Dietary Iron

Dietary iron is available in two forms called heme and non-heme. (**1 mark)**

- **Heme iron** – It is present in fish, meat and chicken. Iron in these sources is found in the form of heme. Heme iron is absorbed easily from intestine. (**2 marks)**

- **Non-heme iron** - Iron in the form of non-heme is available in vegetables, grains and cereals. Non-heme iron is not absorbed easily as heme iron. Cereals, flours and products of grains which are enriched or fortified with iron become good dietary sources of non-heme iron, particularly for children and women. (**2 marks)**

Absorption of Iron

- Iron is absorbed mainly from the small intestine.
- It is absorbed through the enterocytes by pinocytosis and transported into the blood.
- Bile is essential for the absorption of iron. Iron is present mostly in ferric (Fe3+) form. It is converted into ferrous form (Fe2+) which is absorbed into the blood.
- Hydrochloric acid from gastric juice makes the ferrous iron soluble so that it could be converted into ferric iron by the enzyme ferric reductase from enterocytes.
- From enterocytes, ferric iron is transported into blood by a protein called ferroportin.
- In the blood, ferric iron is converted into ferrous iron and transported. (**3 marks – ½ mark for each point)**

Transport of Iron

- After absorption into blood, iron combines with a β-globulin called apotransferrin resulting in the formation of transferrin.
- Iron is transported in blood in the form of transferrin. Iron combines loosely with globin and can be released easily at any region of the body. (**1 mark- ½ mark for each point)**

Storage of Iron

- Iron is stored in large quantities in reticuloendothelial cells and liver hepatocytes
- In the cytoplasm of the cell, iron is stored as ferritin in large amount and Small quantity of iron is also stored as hemosiderin. (**1 mark- ½ mark for each point)**